Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/22/2005
PI Name:	Murad, Ferid M.D., Ph.D.		
Project Title:	A Soluble Guanylyl Cyclase Mouse Knock-Out Model		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ferid.murad@uth.tmc.edu	**Fax:**	FY 713-500-0790
PI Organization Type:	UNIVERSITY	**Phone:**	713-500-7547
Organization Name:	The University of Texas Health Science Center at Houston		
PI Address 1:	6431 Fannin MSB 4098		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77030	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	04/01/2002	**End Date:**	04/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-CA00213		
Performance Goal No.:			
Performance Goal Text:			
	The goal of the proposed study is to generate a soluble guanylyl cyclase (sGC) knockout mouse model to elucidate the role of sGC in the cardiovascular system under physiological and microgravity conditions. While the role for sGC in the vasculature was first identified 20 years ago, the precise understanding of the physiological significance of sGC-cGMP signaling in the myocardium remains incomplete. The exposure of astronauts to microgravity conditions in space results in cardiovascular alterations that are characterized by orthostatic intolerance and decreased exercised capacity, mainly after their return to Earth and gravity. However, the mechanisms of adaptation to microgravity conditions are unknown. As nitric oxide and sGC play a significant role in the cardiovascular system, we hypothesize that sGC plays a role in adaptation to microgravity. To investigate this hypothesis, we will use a gene targeted mouse animal model with myocardium-specific and smooth muscle-specific disruption of gene expression to study the specific role of sGC in microgravity/gravity adaptation. In specific aim 1, an mouse animal model with myocardium-specific and vascular smooth muscle-specific disruption of sGC beta 1 gene will be developed. Following development, we will characterize		

Task Description:	the effects of myocardial-specific and vascular smooth muscle-specific gene disruption on general appearance, embryonic development, histology of the heart and hemodynamic changes in knockout mice versus wild-type mice. In specific aim 2, we will determine the role of sGC pathway deficiency on the development of orthostatic intolerance that occurs during re-adaptation to gravity using the established tail-suspended rodent model to stimulate the microgravity conditions. The time course of changes in cardiovascular function induced by re-adaptation to gravity in knockout mice versus wild type mice will be studied. Finally, the cardiovascular responsiveness in knockout mice versus wild-type mice following de-suspension will be assessed using vaso- and cardio-active agents. Comparisons between knockout mice and wild type mice in simulating weightlessness conditions will aid to identify sGC-dependent pathways necessary for adaptation to microgravity and re-adaptation to gravity. In order to fulfill specific aim 1, we are developing a mouse animal model with myocardium-specific and vascular smooth muscle-specific disruption of sGC gene. During last year we used Lox-Beta 1 sGC targeted vector generated previous year to produce the gene-targeted mouse ES line. Targeted ES cells were introduced into mouse blastocysts and chimeric male mice were obtained. In order to identify animals with a transmition of the targeted ES cells into the germ line chimeric males were bred with wild type females. Presently, we are at the stage of genotyping F1 offsprings to identify Lox-targeted heterozygous mice. Homozygotes with Lox-targeted Beta1 sGC gene will be obtained by crossing identified heterozygotic mice. In the upcoming year, Beta1 sGC-lox mice will be bred with alpha-MyHC-Cre mice containing a myocardium specific Cre-recombinase, to generate myocardium specific Beta1-sGC-Cre/lox knockout mice and [SM-CreER(T2)(ki)] containing smooth muscle-specific Cre recombinase to produce smooth muscle-specific Beta1-sGC-Cre/lox knockout mice. Following their development, we will characterize the effects of myocardial-specific and vascular smooth muscle-specific gene disruption on general appearance, embryonic development, histology of the heart and hemodynamic changes in knockout mice versus wild-type mice.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	sGC is the heterodimeric enzyme that catalyzes the production of cGMP from GTP. In vivo, sGC represents a major receptor for nitric oxide (NO), a unique signaling molecule with numerous physiological functions, including relaxation of smooth muscle, inhibition of platelet aggregation and modulation of cellular differentiation. The cardiovascular functions of sGC that are mediated via activation by NO, together with the particulate isoform of guanylyl cyclase, activated by atrial natriuretic peptide (ANP), are important in the regulation of cardiovascular and renal function. The NO-sGC pathway affects cardiac performance by influencing myocardial contractility, chronotropy and energy production. cGMP produced by vascular smooth muscle sGC in response to NO generated by the vascular endothelium is a major component of vasodilatory signaling pathways in coronary arteries of the heart and other vessels. While the role of sGC in the vasculature was first identified 20 years ago, the physiological significance of sGC-cGMP signaling in the myocardium is not yet fully understood. Indeed, the significance of cGMP signaling in cardiomyocytes is controversial with conflicting reports in the literature. Several problems are associated with the study of sGC-mediated physiology. Distinctions between cGMP-dependent and -independent actions of NO and between the physiological contributions of cGMP produced by the particulate and soluble forms of the enzyme are difficult to determine due to the lack of selective sGC inhibitors. Furthermore, at least two sGC isoforms demonstrate very similar pharmacological and functional properties. A more precise understanding of sGC-cGMP signaling and regulatory events would have profound pathophysiological significance and could potentially help to develop novel therapeutic strategies with minimized negative side effects. A gene-targeted mouse model represents a unique opportunity to study sGC-cGMP action and to clarify the role of sGC in these processes.
Task Progress:	To achieve our goal we developed Lox-Beta1 sGC targeted vector and used it to produce gene-targeted mouse Embryonic Stem cell (ES) line. Targeted ES cells were introduced into mouse blastocysts and chimeric male mice were obtained. To identify animals with transmission of the targeted ES cells into the germ line, chimeric males were bred with wild type C57Bl/6 females. Lox-targeted heterozygous mice were crossed to obtain homozygous mice with Lox-targeted Beta1 sGC gene. Presently, our laboratory maintains the colony of homozygous mice of Beta1-sGC-Lox strain at the Animal Facility of the Institute of Molecular Medicine. The functional targeting of the sGC Beta1 gene in the strain Beta1-sGC-Lox (i.e. that the gene can be eliminated) was confirmed by mating of the Beta1-Lox strain with the Cre strain uniformly expressing Cre recombinase (EIIa-Cre).
Bibliography Type:	Description: (Last Updated: 02/27/2004)
Presentation	Sharina I. G., Doursout M.-F., Murad F. "Soluble guanylyl cyclase mouse knock-out model " Jan-2004